

## Model Development Phase Template

|               |                                 |
|---------------|---------------------------------|
| Date          | 24 June 2026                    |
| Team ID       | SWTID-2026-8296                 |
| Project Title | Credit Card Approval Prediction |
| Maximum Marks | 5 Marks                         |

### Feature Selection Report Template

In the forthcoming update, each feature will be accompanied by a brief description. Users will indicate whether it's selected or not, providing reasoning for their decision. This process will streamline decision-making and enhance transparency in feature selection.

| Feature         | Description                           | Selected (Yes/No) | Reasoning                                                                                      |
|-----------------|---------------------------------------|-------------------|------------------------------------------------------------------------------------------------|
| CODE_GENDER     | Gender of the applicant (Male/Female) | Yes               | Gender can influence risk profile and income stability patterns in some scoring models.        |
| FLAG_OWN_CAR    | Whether the applicant owns a car      | Yes               | Ownership of assets like a car may indicate financial stability and creditworthiness.          |
| FLAG_OWN_REALTY | Whether the applicant owns property   | Yes               | Real estate ownership is a strong indicator of long-term asset value and stability.            |
| CNT_CHILDREN    | Number of children the applicant has  | No                | Dropped to reduce noise; may not directly influence credit approval decisions in this dataset. |

|                     |                                                      |     |                                                                                                 |
|---------------------|------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------|
| AMT_INCOME_TOTAL    | Total annual income of the applicant                 | Yes | A direct indicator of the applicant's repayment ability and financial status.                   |
| NAME_INCOME_TYPE    | Type of income source (e.g., Working, Pensioner)     | Yes | Determines income stability and risk; key feature in risk modeling.                             |
| NAME_EDUCATION_TYPE | Highest education level                              | Yes | Education level often correlates with earning potential and job stability.                      |
| NAME_FAMILY_STATUS  | Marital status of the applicant                      | Yes | Marital status can influence financial responsibilities and household income dynamics.          |
| NAME_HOUSING_TYPE   | Type of housing the applicant lives in               | Yes | Indicates housing security; helps in understanding fixed living costs.                          |
| DAYS_BIRTH          | Age of the applicant (converted from days)           | Yes | Age influences risk—very young or old applicants may have different approval patterns.          |
| DAYS_EMPLOYED       | Duration of employment (in days, converted to years) | Yes | Longer employment generally suggests more stable income and better creditworthiness.            |
| CNT_FAM_MEMBERS     | Total number of family members                       | Yes | May impact disposable income and financial burden; useful for understanding repayment capacity. |
| OCCUPATION_TYPE     | Applicant's job type                                 | No  | Dropped due to many missing values and difficulty in clean                                      |

|                 |                                             |    |                                                                                          |
|-----------------|---------------------------------------------|----|------------------------------------------------------------------------------------------|
|                 |                                             |    | encoding; introduces inconsistency.                                                      |
| FLAG_MOBIL      | Whether the mobile number is provided       | No | Dropped due to constant or near-constant value across all entries; low information gain. |
| FLAG_WORK_PHONE | Whether the applicant provided a work phone | No | Low predictive power and correlation with target variable.                               |
| FLAG_PHONE      | Whether a phone number was provided         | No | Same as above; dropped to reduce noise.                                                  |
| FLAG_EMAIL      | Whether email was provided                  | No |                                                                                          |